

Functions

Functions

Definition 1:

Let A and B be sets. A *function* f from A to B is an assignment of exactly one element of B to each element of A . We write $f(a) = b$ if b is the unique element of B assigned by the function f to the element a of A . If f is a function from A to B , we write $f: A \rightarrow B$.

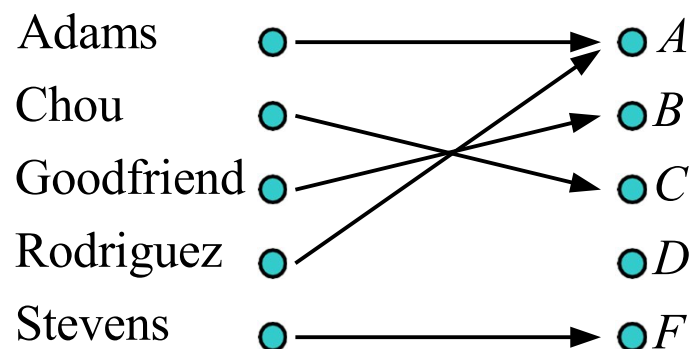


Figure: Assignment of Grades in a Discrete Mathematics Class.

Definition 2:

If f is a function from A to B , we say that A is the *domain* of f and B is the *codomain* of f .

If $f(a) = b$, we say that b is the *image* of a and a is a *pre-image* of b .

The *range* of f is the set of all images of elements of A .

Also, if f is a function from A to B , we say that f maps A to B .

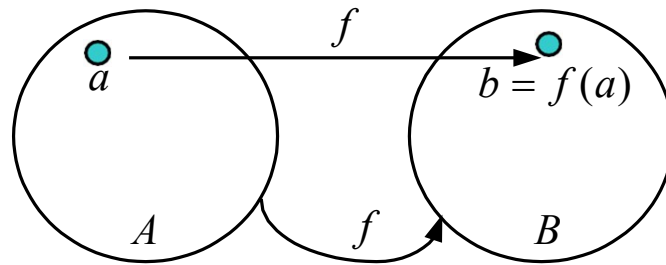
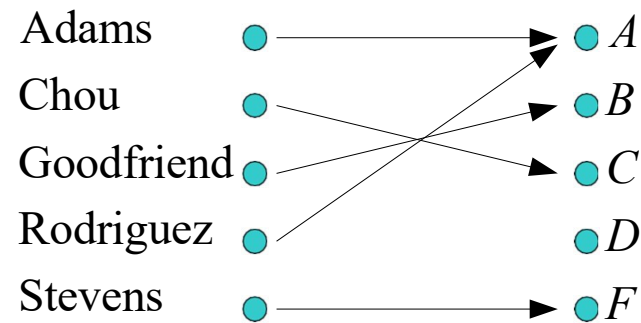


Figure: The function f maps A to B .

Example: What are the domain, codomain, and range of the function that assigns grades to students in discrete mathematics class described in the following figure?



Solution:

Let G be the function that assigns a grade to a student in discrete mathematics class.

$$G(\text{Adams}) = A.$$

The domain of G is the set $\{\text{Adams, Chou, Goodfriend, Rodriguez, Stevens}\}$

The codomain of G is the set $\{A, B, C, D, F\}$

The range of G is the set $\{A, B, C, F\}$.

Example: Let R be the relation consisting of ordered pairs (Abdul, 22), (Brenda, 24), (Carla, 21), (Desire, 22), (Eddie, 24), and (Felicia, 22), where each pair consists of a graduate student and the age of this student. What is the function that this relation determines?

Solution:

This relation defines the function f , where

$$f(\text{Abdul}) = 22$$

$$f(\text{Brenda}) = 24$$

$$f(\text{Carla}) = 21$$

$$f(\text{Desire}) = 22$$

$$f(\text{Eddie}) = 24$$

$$f(\text{Felicia}) = 22.$$

Here the domain is the set

$\{\text{Abdul, Brenda, Carla, Desire, Eddie, Felicia}\}.$

The codomain is the set of positive integers.

The range is the set $\{21, 22, 24\}.$

Example: Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ assign the square of an integer to an integer. Then, $f(x) = x^2$, where the domain of f is the set of all integers, the codomain of f is the set of all integers, and the range of f is the set of all integers that are perfect squares, namely, $\{0, 1, 4, 9, \dots\}$.

Definition 3: Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x),$$

$$(f_1 f_2)(x) = f_1(x) f_2(x).$$

Example: Let f_1 and f_2 be functions from $\mathbf{R}$ to $\mathbf{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution:

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

$$(f_1 f_2)(x) = f_1(x) f_2(x) = x^2 (x - x^2) = x^3 - x^4$$

One-to-One and Onto Functions

Definition 5: A function f is said to be *one-to-one*, or *injective*, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f .

Example: Determine whether the function f from $\{a, b, c, d\}$ to $\{1, 2, 3, 4, 5\}$ with $f(a) = 4$, $f(b) = 5$, $f(c) = 1$, and $f(d) = 3$ is one-to-one.

Solution:

The function f is one-to-one, since f takes on different values at the four elements of its domain.

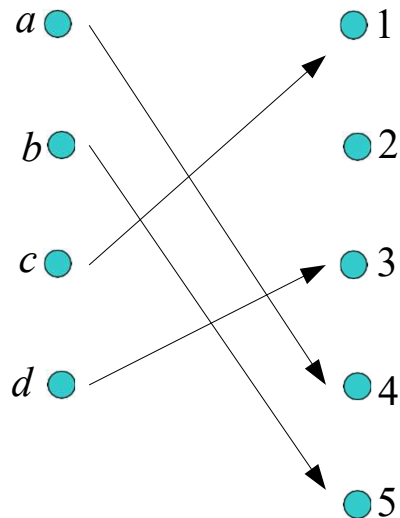


Figure 3 A one-to-one Function.

Example: Determine whether the function $f(x) = x^2$ from the set of integers to the set of integers is one-to-one.

Solution:

The function $f(x) = x^2$ is not one-to-one, since $f(1) = f(-1) = 1$, but $1 \neq -1$.

Example: Determine whether the function $f(x) = x + 1$ from the set of real numbers to itself is one-to-one.

Solution:

$$f(x) = x + 1$$

$$f(y) = y + 1$$

If $f(x) = f(y)$, then $x + 1 = y + 1$.

This implies that $x = y$.

Therefore, the function $f(x) = x + 1$ is one-to-one.

Definition 7: A function f from A to B is called *onto*, or *surjective*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function?

Solution:

Since all three elements of the codomain are images of elements in the domain, f is onto.

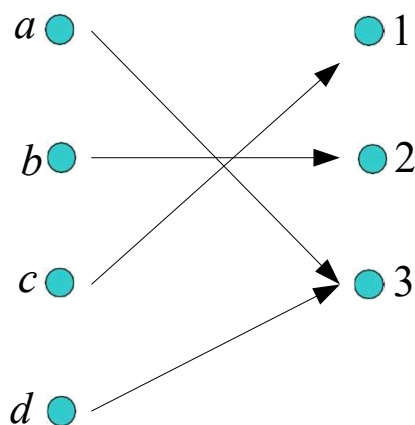


Figure: An Onto Function.

Example: Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

Solution:

The function f is not onto since there is no integer x with $x^2 = -1$, for example.

Example: Is the function $f(x) = x + 1$ from the set of integers to the set of integers onto?

Solution:

This function is onto, since for every integer y from the codomain, there is an integer x in the domain such that $f(x) = y = x + 1$.

Definition 8: The function f is a *one-to-one correspondence*, or a *bijective*, if it is both one-to-one and onto.

Example: Let f be the function from $\{a, b, c, d\}$ to $\{1, 2, 3, 4\}$ with $f(a) = 4$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f a bijective function?

Solution:

$$f(a) = 4$$

$$f(b) = 2$$

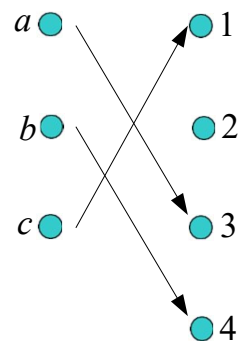
$$f(c) = 1$$

$$f(d) = 3$$

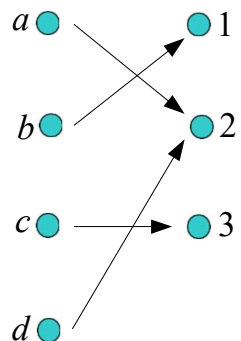
Each element of the domain maps to unique element of the codomain. So, the function f is one-to-one.

All elements of the codomain are mapped. So, the function f is onto.

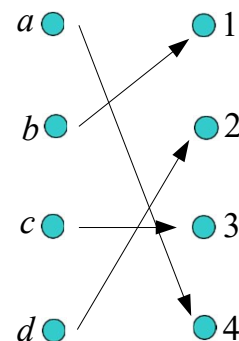
As the function f is both one-to-one and onto function, it is a bijective function.



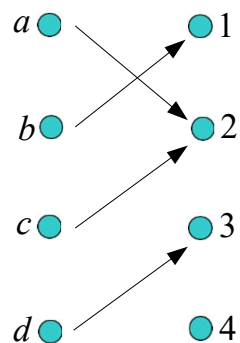
one - to - one,
not onto



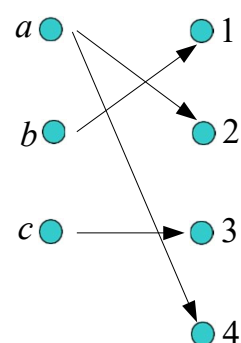
onto,
not one - to - one



one - to - one,
and onto



Neither one - to - one,
nor onto



Not a function

Figure: Examples of Different Types of Correspondance.

Inverse Functions and Compositions of Functions

Definition 9: Let f be a bijective function from the set A to the set B . The *inverse function* of f is the function that assigns to an element b belonging to B the unique element a in A such that $f(a) = b$. The inverse function of f is denoted by f^{-1} . Hence, $f^{-1}(b) = a$ when $f(a) = b$.

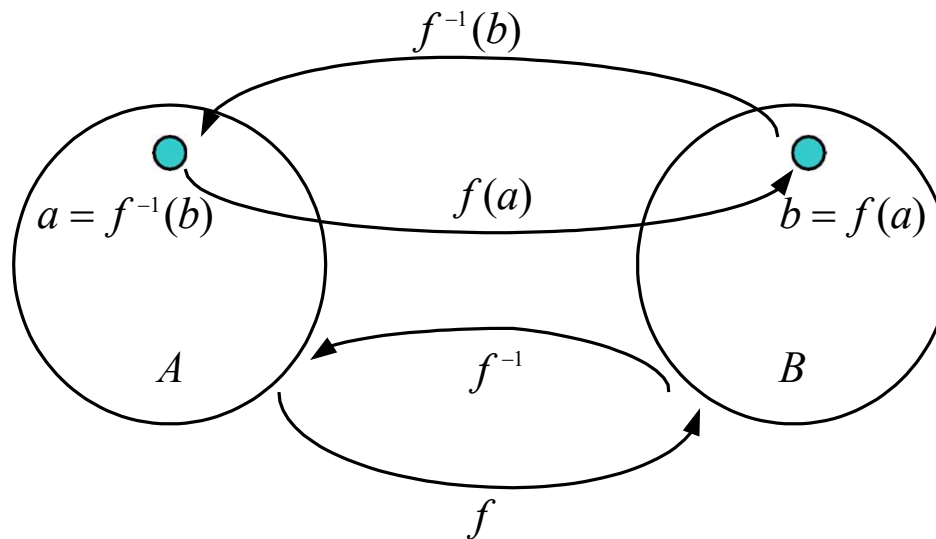


Figure: The Function f^{-1} is the Inverse of Function f .

- If a function f is not bijective, we cannot define an inverse function of f .
- A bijective function is called **invertible** because we can define an inverse of this function.
- A function is **not invertible** if it is not bijective function, because the inverse of such function does not exist.

Example: Let f be the function from $\{a, b, c\}$ to $\{1, 2, 3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible, and if it is, what is its inverse?

Solution:

$$f(a) = 2$$

$$f(b) = 3$$

$$f(c) = 1$$

The function f is one-to-one and onto, so it is a bijective function and is invertible.

The inverse function f^{-1} is

$$f^{-1}(1) = c$$

$$f^{-1}(2) = a$$

$$f^{-1}(3) = b.$$

Example: Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if it is, what is its inverse?

Solution:

The function f is a one-to-one function.

The function f is an onto function.

So, the function f is a bijective function and is invertible.

Suppose that y is the image of x , so that $y = x + 1$.

Then $x = y - 1$. This means that $y - 1$ is the unique element of $\mathbf{Z}$ that is sent to y by f .

Consequently, $f^{-1}(y) = y - 1$.

Example: Let f be the function from $\mathbf{R}$ to $\mathbf{R}$ with $f(x) = x^2$. Is f invertible?

Solution:

Since $f(-2) = f(2) = 4$, f is not one-to-one. So, f is not a bijective function. Hence, f is not invertible.

Example: Let f be the function from the set of all nonnegative real numbers to the set of all nonnegative real numbers with $f(x) = x^2$. Show that f is invertible.

Solution:

If $f(x) = f(y)$, then $x^2 = y^2$, so $x^2 - y^2 = (x + y)(x - y) = 0$.

This means that $x + y = 0$ or $x - y = 0$, so $x = -y$ or $x = y$. Because both x and y are nonnegative, we must have $x = y$. so, $f(x) = x^2$ is one-to-one.

Furthermore, $f(x) = x^2$ is onto when the codomain is the set of all nonnegative real numbers, because each nonnegative real number has a square root. That is, if y is a nonnegative real number, there exists a nonnegative real number x such that $x = \sqrt{y}$, which means that $x^2 = y$.

Because the function $f(x) = x^2$ from the set of nonnegative real numbers to the set of nonnegative real numbers is one-to-one and onto, it is a bijective function and is invertible.

Definition 10: Let g be a function from the set A to the set B and let f be a function from the set B to the set C . The composition of the functions f and g , denoted by $f \circ g$, is defined by $(f \circ g)(a) = f(g(a))$.

To find $(f \circ g)(a)$ we first apply the function g to a to obtain $g(a)$ and then we apply the function f to the result of $g(a)$ to obtain $(f \circ g)(a) = f(g(a))$.

The composition $f \circ g$ cannot be defined unless the range of g is a subset of the domain of f .

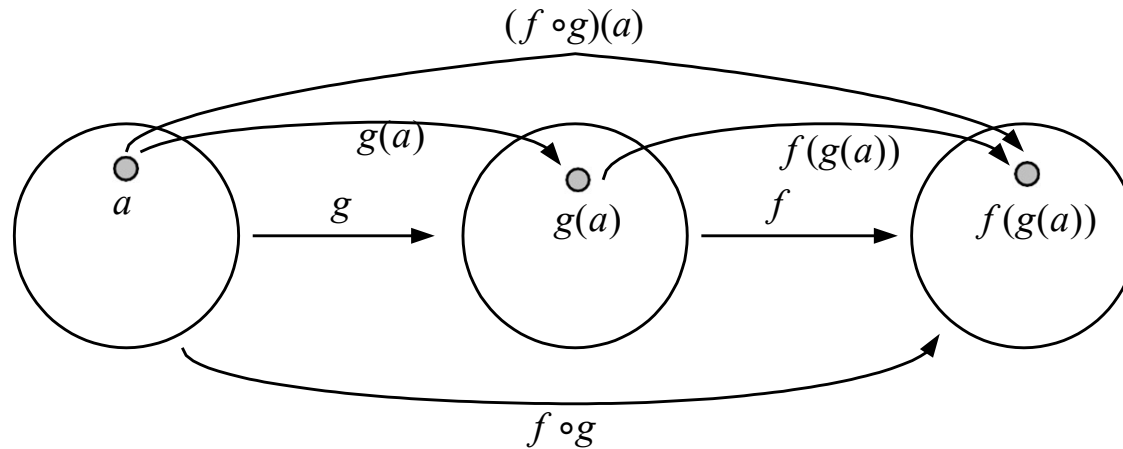


Figure: The Composition of the Functions f and g .

Example: Let g be the function from the set $\{a, b, c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a, b, c\}$ to the set $\{1, 2, 3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$. What is the composition of f and g , and what is the composition of g and f ?

Solution:

$$g(a) = b$$

$$g(b) = c$$

$$g(c) = a$$

$$f(a) = 3$$

$$f(b) = 2$$

$$f(c) = 1$$

The composition $f \circ g$ is defined by

$$(f \circ g)(a) = f(g(a)) = f(b) = 2,$$

$$(f \circ g)(b) = f(g(b)) = f(c) = 1,$$

$$(f \circ g)(c) = f(g(c)) = f(a) = 3.$$

$g \circ f$ is not defined, because the range of f is not a subset of the domain of g .

Example: Let f and g be the functions from the set of integers to the set of integers defined by $f(x) = 2x + 3$ and $g(x) = 3x + 2$. What is the composition of f and g ? What is the composition of g and f ?

Solution:

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11$$

Recursive Function

Recursively Defined Functions

We use two steps to define a function with the set of nonnegative integers as its domain:

BASIS STEP: Specify the value of the function at zero.

RECURSIVE STEP: Derive a rule for finding its value at an integer from its values at smaller integers.

Such a definition is called a **recursive** or **inductive** definition.

Example: Suppose that f is defined recursively by

$$f(0) = 3$$

$$f(n) = 2f(n-1) + 3 \text{ for } n = 1, 2, 3, \dots$$

Find $f(1)$, $f(2)$, $f(3)$, and $f(4)$.

Solution:

$$f(0) = 3$$

$$f(1) = 2f(0) + 3 = 2 \cdot 3 + 3 = 9$$

$$f(2) = 2f(1) + 3 = 2 \cdot 9 + 3 = 21$$

$$f(3) = 2f(2) + 3 = 2 \cdot 21 + 3 = 45$$

$$f(4) = 2f(3) + 3 = 2 \cdot 45 + 3 = 93$$

Example: Give a recursive definition of the factorial function $F(n) = n!$.

Solution:

$$F(0) = 1$$

$$F(n) = n \cdot F(n-1) \text{ for } n = 1, 2, 3, \dots$$

$$F(5) = 5 \cdot F(4)$$

$$= 5 \cdot 4 \cdot F(3)$$

$$= 5 \cdot 4 \cdot 3 \cdot F(2)$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot F(1)$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot F(0)$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 1$$

$$= 120$$

Example: Give a recursive definition of a^n , where a is a nonzero real number and n is a nonnegative integer.

Solution:

$$a^0 = 1$$

$$a^n = a \cdot a^{n-1} \text{ for } n = 1, 2, 3, \dots$$

Example: The *Fibonacci numbers*, $f_0, f_1, f_2, \dots$, are defined by the equations

$$f_0 = 0$$

$$f_1 = 1$$

$$f_n = f_{n-1} + f_{n-2} \text{ for } n = 2, 3, 4, \dots$$

Find the Fibonacci numbers f_2, f_3, f_4, f_5 , and f_6 .

Solution:

$$f_0 = 0$$

$$f_1 = 1$$

$$f_2 = f_1 + f_0 = 1 + 0 = 1$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8$$